



# Asthma

Synthetic reference

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## Identity

|                         |                                                     |
|-------------------------|-----------------------------------------------------|
| Model                   | Asthma                                              |
| Reference case          | Synthetic reference fixture                         |
| Reference country       | XSC                                                 |
| Reference parameter set | reference-parameter-set:ncd-asthma:synthetic-v1     |
| Scenario registry       | scenario-registry:ncd-asthma:reference-synthetic-v1 |
| Source model hash       | 94c67579de128117                                    |

## Reference Case

|                |                             |
|----------------|-----------------------------|
| Output years   | 2000-2100                   |
| Engine runtime | 2000-2101                   |
| Data bundle    | 552cbb7f0cfebb94            |
| Bundle mode    | synthetic_reference_fixture |

## Fixture Provenance

| Parameter                             | Reference value                                                 | Source                                                    |
|---------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| Population                            | WHO standard age distribution, total 1,000,000                  | Forecast Health deterministic synthetic reference fixture |
| Asthma epidemiology                   | Synthetic low-rate disease fixture                              | Forecast Health deterministic synthetic reference fixture |
| Demographic rates                     | Synthetic fertility, mortality, migration, and sex-ratio arrays | Forecast Health deterministic synthetic reference fixture |
| Risk factors and intervention effects | Neutral or low-rate synthetic fixture values                    | Forecast Health deterministic synthetic reference fixture |
| Economic and burden inputs            | Synthetic GDP, labour-force, disability, and YLL values         | Forecast Health deterministic synthetic reference fixture |

## Internal Validity

|                         |                             |
|-------------------------|-----------------------------|
| Synthetic reference run | pass                        |
| Result rows             | 66803                       |
| Data bundle             | synthetic_reference_fixture |
| Bundle inputs           | 1230                        |
| Headline checks         | pass                        |
| Result hash             | 06fdce1535958f27            |

## Headline Outputs

| Year | Asthma prevalence state | Asthma incidence flow | Asthma mortality flow |
|------|-------------------------|-----------------------|-----------------------|
| 2000 | 52,293                  | 4,745                 | 5                     |
| 2001 | 54,460                  | 4,733                 | 5                     |
| 2002 | 56,511                  | 4,722                 | 5                     |
| 2003 | 58,450                  | 4,712                 | 6                     |
| 2004 | 60,284                  | 4,702                 | 6                     |
| 2005 | 62,019                  | 4,693                 | 6                     |
| 2006 | 63,645                  | 4,684                 | 6                     |
| 2007 | 65,182                  | 4,674                 | 6                     |
| 2008 | 66,635                  | 4,665                 | 6                     |
| 2009 | 68,007                  | 4,657                 | 7                     |
| 2010 | 69,303                  | 4,649                 | 7                     |
| 2011 | 70,487                  | 4,641                 | 7                     |
| 2012 | 71,601                  | 4,630                 | 7                     |
| 2013 | 72,650                  | 4,621                 | 7                     |
| 2014 | 73,637                  | 4,611                 | 7                     |
| 2015 | 74,566                  | 4,602                 | 7                     |
| 2016 | 75,368                  | 4,593                 | 7                     |
| 2017 | 76,116                  | 4,580                 | 7                     |
| 2018 | 76,813                  | 4,567                 | 7                     |
| 2019 | 77,462                  | 4,555                 | 8                     |
| 2020 | 78,066                  | 4,542                 | 8                     |
| 2021 | 78,529                  | 4,530                 | 8                     |
| 2022 | 78,950                  | 4,513                 | 8                     |
| 2023 | 79,331                  | 4,496                 | 8                     |
| 2024 | 79,674                  | 4,478                 | 8                     |
| 2025 | 79,982                  | 4,462                 | 8                     |
| 2026 | 80,142                  | 4,445                 | 8                     |
| 2027 | 80,269                  | 4,422                 | 8                     |
| 2028 | 80,365                  | 4,399                 | 8                     |
| 2029 | 80,432                  | 4,377                 | 8                     |
| 2030 | 80,471                  | 4,354                 | 8                     |
| 2031 | 80,356                  | 4,332                 | 8                     |
| 2032 | 80,215                  | 4,303                 | 8                     |
| 2033 | 80,050                  | 4,274                 | 8                     |
| 2034 | 79,863                  | 4,246                 | 8                     |
| 2035 | 79,654                  | 4,217                 | 8                     |
| 2036 | 79,293                  | 4,188                 | 8                     |
| 2037 | 78,911                  | 4,153                 | 8                     |
| 2038 | 78,511                  | 4,118                 | 8                     |
| 2039 | 78,094                  | 4,083                 | 8                     |
| 2040 | 77,660                  | 4,047                 | 8                     |
| 2041 | 77,064                  | 4,012                 | 8                     |
| 2042 | 76,452                  | 3,970                 | 8                     |
| 2043 | 75,826                  | 3,928                 | 7                     |
| 2044 | 75,187                  | 3,885                 | 7                     |
| 2045 | 74,534                  | 3,843                 | 7                     |

| Year | Asthma prevalence state | Asthma incidence flow | Asthma mortality flow |
|------|-------------------------|-----------------------|-----------------------|
| 2046 | 73,724                  | 3,801                 | 7                     |
| 2047 | 72,901                  | 3,751                 | 7                     |
| 2048 | 72,068                  | 3,701                 | 7                     |
| 2049 | 71,224                  | 3,652                 | 7                     |
| 2050 | 70,371                  | 3,602                 | 7                     |
| 2051 | 69,387                  | 3,553                 | 7                     |
| 2052 | 68,395                  | 3,498                 | 7                     |
| 2053 | 67,395                  | 3,442                 | 7                     |
| 2054 | 66,387                  | 3,387                 | 7                     |
| 2055 | 65,371                  | 3,332                 | 6                     |
| 2056 | 64,249                  | 3,276                 | 6                     |
| 2057 | 63,120                  | 3,216                 | 6                     |
| 2058 | 61,986                  | 3,156                 | 6                     |
| 2059 | 60,845                  | 3,096                 | 6                     |
| 2060 | 59,699                  | 3,035                 | 6                     |
| 2061 | 58,446                  | 2,975                 | 6                     |
| 2062 | 57,188                  | 2,910                 | 6                     |
| 2063 | 55,926                  | 2,845                 | 6                     |
| 2064 | 54,659                  | 2,780                 | 5                     |
| 2065 | 53,389                  | 2,715                 | 5                     |
| 2066 | 52,030                  | 2,649                 | 5                     |
| 2067 | 50,668                  | 2,580                 | 5                     |
| 2068 | 49,302                  | 2,511                 | 5                     |
| 2069 | 47,934                  | 2,442                 | 5                     |
| 2070 | 46,563                  | 2,373                 | 5                     |
| 2071 | 45,130                  | 2,303                 | 5                     |
| 2072 | 43,694                  | 2,231                 | 4                     |
| 2073 | 42,257                  | 2,159                 | 4                     |
| 2074 | 40,817                  | 2,087                 | 4                     |
| 2075 | 39,375                  | 2,015                 | 4                     |
| 2076 | 37,877                  | 1,943                 | 4                     |
| 2077 | 36,378                  | 1,868                 | 4                     |
| 2078 | 34,877                  | 1,794                 | 4                     |
| 2079 | 33,375                  | 1,719                 | 3                     |
| 2080 | 31,871                  | 1,645                 | 3                     |
| 2081 | 30,320                  | 1,570                 | 3                     |
| 2082 | 28,767                  | 1,493                 | 3                     |
| 2083 | 27,214                  | 1,416                 | 3                     |
| 2084 | 25,659                  | 1,339                 | 3                     |
| 2085 | 24,103                  | 1,262                 | 3                     |
| 2086 | 22,522                  | 1,186                 | 2                     |
| 2087 | 20,941                  | 1,108                 | 2                     |
| 2088 | 19,358                  | 1,030                 | 2                     |
| 2089 | 17,775                  | 952                   | 2                     |
| 2090 | 16,191                  | 874                   | 2                     |
| 2091 | 14,590                  | 796                   | 2                     |
| 2092 | 12,988                  | 717                   | 1                     |
| 2093 | 11,386                  | 638                   | 1                     |



| Year | Asthma prevalence state | Asthma incidence flow | Asthma mortality flow |
|------|-------------------------|-----------------------|-----------------------|
| 2094 | 9,783                   | 559                   | 1                     |
| 2095 | 8,179                   | 480                   | 1                     |
| 2096 | 6,544                   | 402                   | 1                     |
| 2097 | 4,909                   | 321                   | 1                     |
| 2098 | 3,273                   | 241                   | 0                     |
| 2099 | 1,637                   | 161                   | 0                     |
| 2100 | 0                       | 80                    | 0                     |